
Please read the following instructions carefully:

- Write your **name, roll number and course code** on top of the sheet.
 - Write your answers in **neat and clear** handwriting!
 - **Do not waste paper** in unnecessary embellishments/top sheets; write on both sides of the paper. It is only the content of your assignment sheet that carries marks!
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1. Construct a CFG that recognizes the language L_1 where
 $L_1 = \{w | w \text{ starts and ends with the same symbol and } |w| \text{ is odd}\}$
Convert the grammar to Chomsky Normal Form. [6+4]
2. Construct a Nondeterministic PDA that accepts the CFL L_2 , where
 $L_2 = \{w | w \text{ has equal numbers of 0s and 1s}\}$
Explain how the NPDA works, in your own words using (i) a string in the language and (ii) a string not in the language. [6+4]
3. Prove that the language L_3 is not context-free
 $L_3 = \{1^n 2^n 3^m 4^m | n, m \geq 1\}$ [5]